AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre SSC-II General Mathematics Annual Examinations 2023

Introduction

This document has been prepared for the teachers and candidates of Secondary School Certificate (SSC) Part II (Class X) General Mathematics. It contains comments on candidates' responses to the 2023 SSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Candidates performed really well in some concepts, such as, Personal Income, Arithmetic Mean, Square Root by long division, Linear Equations and Arithmetic Series. However, candidates who did not score well mostly failed to understand the demands of the questions, often misinterpreting the command words and the stimuli.

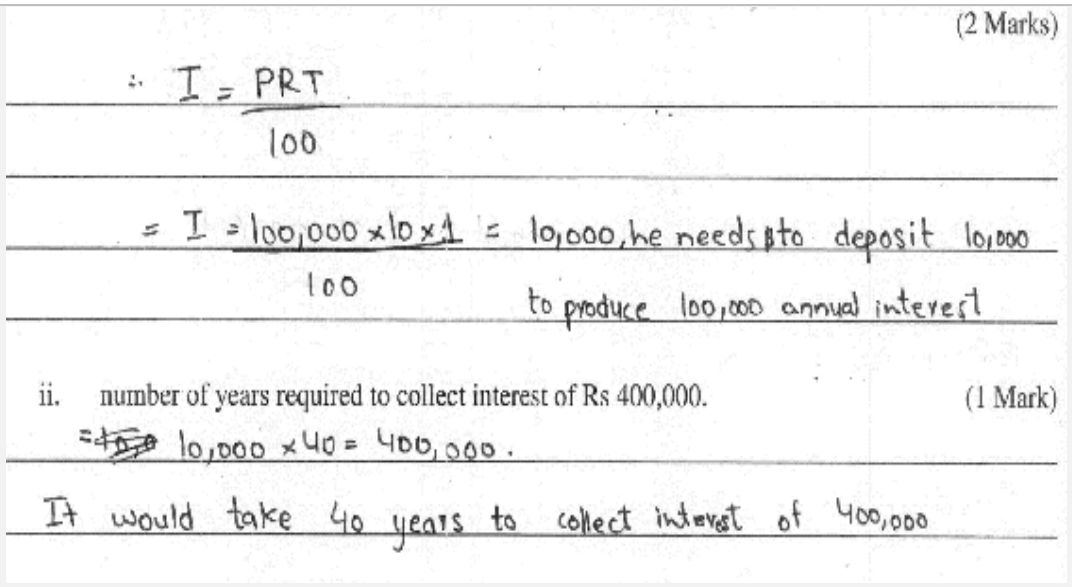
Mentioned below are few concepts that teachers need to focus so that the candidates may perform better.

- Simple Interest
- HCF and LCM by division method
- Quadratic Equation
- Fundamentals of Geometry
- Area Volume

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

Question No. 1a Candidates were given the choice to attempt any ONE out of the two questions: 1a and 1b.	
Question Text	Akhtar intends to deposit some money in his bank account. The bank offers 10% simple interest rate annually. Assuming that the interest is drawn on yearly basis, calculate the i. amount of money he needs to deposit if he aims to produce Rs 100,000 annual interest. ii. number of years required to collect interest of Rs 400,000.
SLO No.	11.3.2
SLO Text	Solve problems related to simple interest.
Max Marks	3
Cognitive Level	A*
Checking Hints	i. 1 mark for substituting the values. 1 mark for calculating the principal amount. ii. 1 mark for calculating number of years. (**Note**: Mark would be granted if candidates directly write t = 4 years.)
Overall Performance	Lesser number of candidates opted for part 'a'. In such responses, candidates correctly identified the concept of simple interest and solved the question accurately by applying the appropriate formula. However, a significant number of candidates faced difficulties in substitution the given values in the formula correctly.
Description of Better Responses	Better responses showed clear understanding of the problem, i.e., correct identification of the information given in question, correct for substitution in the simple interest formula and correct calculations. They showed the working of all the required.
Image of Better Response	$I = \frac{PRT}{100}$ $\quad 100,000 \times 100 = 10x$ $10,000,000 = 10x$ $100,000 = \frac{x(10)(1)}{100}$ $\quad \frac{10,000,000}{10} = x$ $x = Rs\ 1,000,000$ ii. number of years required to collect interest of Rs 400,000. (1 Mark) $I = PRT/100$ $\quad 400,000 = \frac{1,000,000(10)(x}{100}$ $400,000 = 100,000x$ $\quad 400,000 \div 100,000 = x$ $\quad x = 4$ years
Description of Weaker Responses	Weaker responses showed insufficient understanding of the problem. The information given in the question was not substituted correctly in the formula. In addition, some candidates showed calculation errors.

Image of Weaker Response	

(2 Marks)

$\therefore I = \frac{PRT}{100}$

$= I = \frac{100,000 \times 10 \times 1}{100} = 10,000$, he needs to deposit 10,000 to produce 100,000 annual interest

ii. number of years required to collect interest of Rs 400,000. (1 Mark)

$= 10,000 \times 40 = 400,000$.

It would take 40 years to collect interest of 400,000

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy** Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
Teachers may implement these teaching methodologies to develop a stronger foundation in simple interest.
Case Studies: Provide students with case studies involving real individuals or businesses dealing with simple interest calculations. Ask students to analyse these cases and propose solutions.
Current Events Analysis: Have students analyse current news articles or reports related to interest rates and financial markets. Encourage them to discuss how these events impact simple interest calculations.
Field Trips: Organise field trips to financial institutions or banks where students can witness first-hand how interest rates and financial transactions work.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

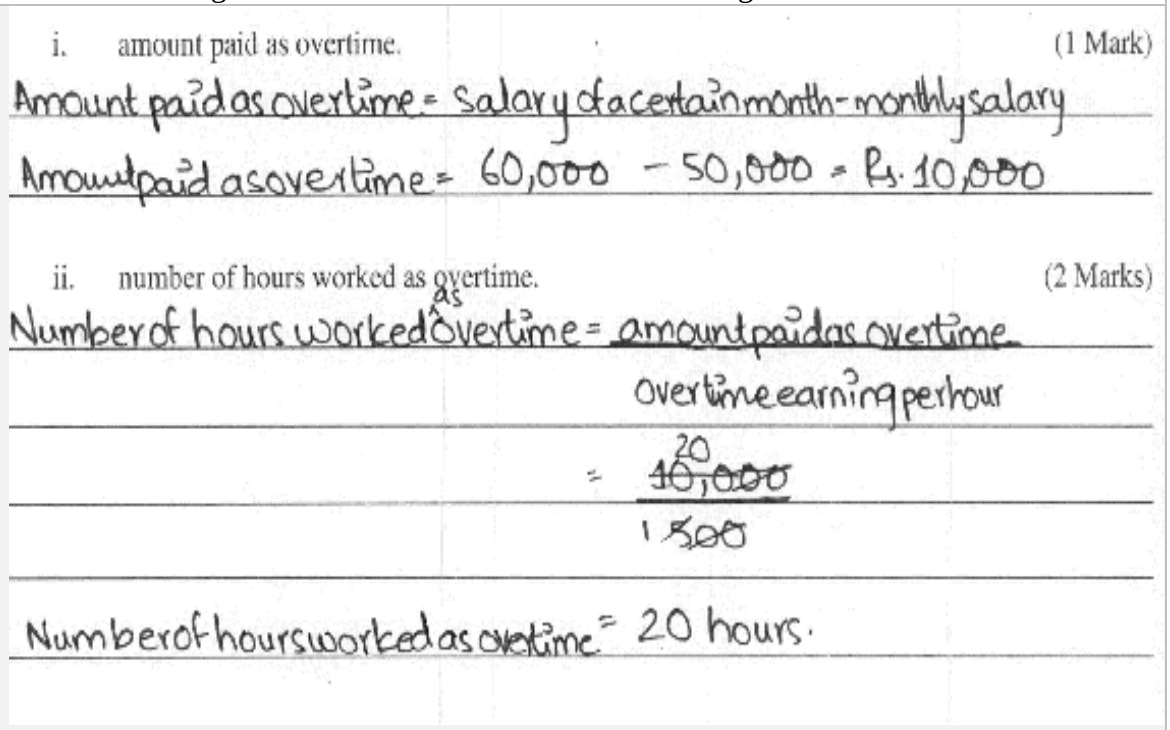
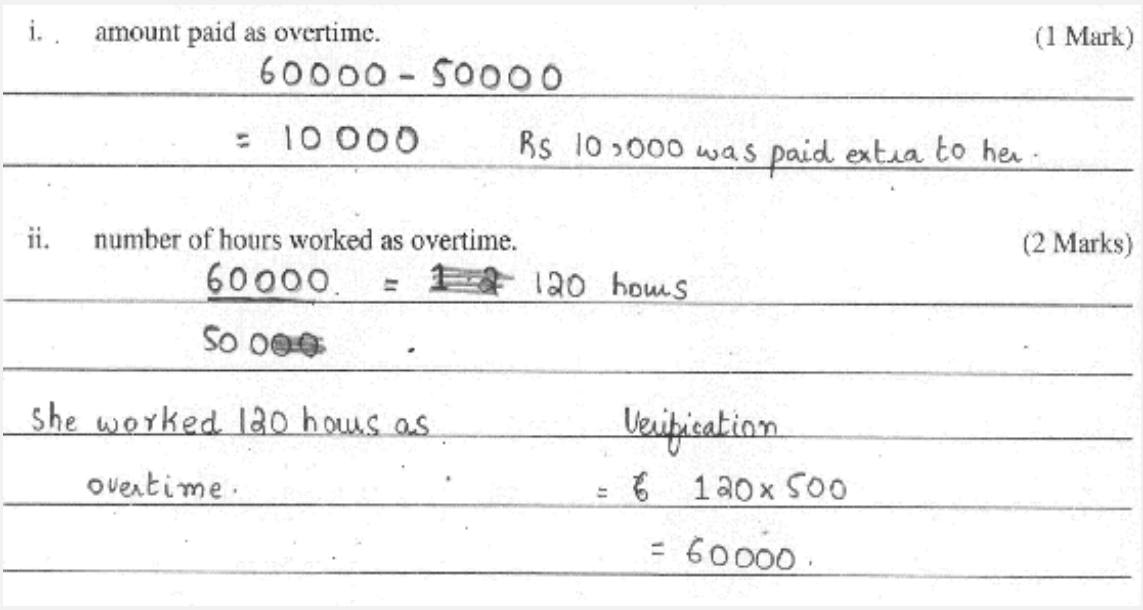
Question No. 1b Candidates were given the choice to attempt any ONE out of the two questions: 1a and 1b.	
Question Text	Ayesha works in a private company where she is paid a monthly salary of Rs 50,000. For working overtime, she is paid Rs 500 per hour. In a certain month, she was paid Rs 60,000. Calculate the i. amount paid as overtime. ii. number of hours worked as overtime.
SLO No.	12.3.1b
SLO Text	Calculate personal income (weekly, monthly and annually) of b) a worker who is paid for overtime on hourly basis in addition to his daily wages.
Max Marks	3
Cognitive Level	A
Checking Hints	i. 1 mark for calculating payment for overtime ii. 1 mark for writing $10{,}000 = n \times 500$ 1 mark for calculating the number of overtime hour (**Note**: 1 mark would be granted if candidate did not show $10{,}000 = n \times 500$ and directly calculated number of overtime hours.)
Overall Performance	Majority of the candidates count attempted this question and most of them obtained full marks. Thus, this part proved easier for the candidates.
Description of Better Responses	Better responses demonstrated clear understanding of the problem, proper working and correct calculations. Candidates reflected comprehensive understanding of the concept of overtime earning with reference to the number of working hours.
Image of Better Response	 i. amount paid as overtime. (1 Mark) Amount paid as overtime = Salary of a certain month - monthly salary Amount paid as overtime = 60,000 − 50,000 = Rs. 10,000 ii. number of hours worked as overtime. (2 Marks) Number of hours worked as overtime = amount paid as overtime / overtime earning per hour = 10,000 / 500 = 20 Number of hours worked as overtime = 20 hours.
Description of Weaker Responses	Weaker responses demonstrated weaker understanding of the concept of overtime payments and its relationship with time (number of hours). Some students attempted part (i) correctly, however, for part (ii), they applied the calculation on the total earning instead of overtime earning.

Image of Weaker Response	

Suggestions for Improvement (Highlighted part)

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• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

Teachers may implement these teaching methodologies to develop a stronger foundation in simple interest Case Studies: Provide case studies of different workers with varying overtime rates, daily wages, and weekly work hours. Have students analyse these cases, calculate personal incomes, and discuss their results as a class.

Interactive Calculations: Use a whiteboard or interactive software to perform sample calculations in real-time. Encourage students to participate and provide input as you calculate the worker's income for a week, month, and year.

Role-Playing: Organise a role-playing activity where students take on the roles of both the worker and the employer. They can negotiate daily wages, overtime rates, and work hours to gain a better understanding of how these factors affect personal income.
Real Data Analysis: Provide real data from payslips or salary information (with personal details removed) and ask students to analyse and calculate personal incomes based on the provided data. This can make the concept more tangible.

<table>
<tr><th colspan="2">Question No. 2</th></tr>
<tr><td>Question Text</td><td>The given table shows the number of daily mails received during a particular week in an office. Complete the table and calculate the arithmetic mean.
<table>
<tr><th>Number of mails</th><th>Number of occurrences (f)</th><th>Class Mark (x)</th><th>fx</th></tr>
<tr><td>11-15</td><td>12</td><td></td><td></td></tr>
<tr><td>16-20</td><td>10</td><td></td><td></td></tr>
<tr><td>21-25</td><td>25</td><td></td><td></td></tr>
<tr><td>26-30</td><td>19</td><td></td><td></td></tr>
<tr><td>31-35</td><td>9</td><td></td><td></td></tr>
<tr><td>Total</td><td>75</td><td></td><td></td></tr>
</table></td></tr>
<tr><td>SLO No.</td><td>13.3.1</td></tr>
<tr><td>SLO Text</td><td>Calculate arithmetic mean, median and mode for ungrouped and grouped data.</td></tr>
<tr><td>Max Marks</td><td>3</td></tr>
<tr><td>Cognitive Level</td><td>A</td></tr>
<tr><td>Checking Hints</td><td>i. 1 mark for filling the column of class marks
1 mark for calculating fx
ii. 1 mark for substituting the values in the formula for AM
1 mark for calculating AM
(Note: marks would not be deducted for less than 4 mistakes in filling the columns)</td></tr>
<tr><td>Overall Performance</td><td>In this question, the majority of candidates encountered difficulty in completing the table and applying the correct mean formula. A common mistake was calculating cumulative frequency, which was not required for this particular problem. Additionally, some candidates used the mean formula for ungrouped data, which was not applicable in this context. Overall, only a few candidates successfully achieved full marks on this question, while others achieved partial correctness in their responses.</td></tr>
<tr><td>Description of Better Responses</td><td>Better responses exhibited clear understanding of the required concept and completed the table by finding class mark and frequency. They then found arithmetic mean by applying correct formula of arithmetic mean.</td></tr>
</table>

Image of Better Response

Number of mails	Number of occurrences (f)	Class Mark (x)	fx
11-15	12	$\frac{11+15}{2} = 13$	156
16-20	10	$\frac{16+20}{2} = 18$	180
21-25	25	$\frac{21+25}{2} = 23$	575
26-30	19	$\frac{26+30}{2} = 28$	532
31-35	9	$\frac{31+35}{2} = 33$	297
Total	75	115	1740

formula $= \frac{\Sigma fx}{\Sigma f} = \frac{1740}{75} = 23.2$

Description of Weaker Responses

Weaker responses included inappropriate completion of the table, most candidates found the cumulative frequency in the column of $f(x)$, whereas some candidate calculated class boundaries instead of class marks. This was followed by application of incorrect formula or incorrect substitution in the correct formula. In addition, there were candidates who made the correct substitution in the correct formula but made calculation mistakes in finding the answer.

Images of Weaker Response

Image (i)

Number of mails	Number of occurrences (f)	Class Mark (x)	fx
11-15	12	3	36
16-20	10	2.5	25
21-25	25	6.25	156.25
26-30	19	4.75	90.25
31-35	9	2.25	20.25
Total	75	37.5	327.75

AM $= \frac{a+b}{2}$

i) $= \frac{12+3}{2} = 7.5$

ii) $= \frac{10+2.5}{2} = 6.25$

iii) $= \frac{25+6.25}{2} = 15.625$

iv) $= \frac{9+2.25}{2} = 11.25$

Total $= \frac{75+37.5}{2}$

$= 56.25$

Image (ii)

Number of mails	Number of occurrences (f)	Class Mark (x)	fx
11-15	12	10.5 - 15.5	12
16-20	10	15.5 - 20.5	22
21-25	25	20.5 - 25.5	47
26-30	19	25.5 - 30.5	66
31-35	9	30.5 - 35.5	75
Total	75	102.5 - 127.5	222

The formula of mean is $\bar{x} = \frac{\Sigma fx}{N}$

$\frac{222}{75} = 2.96$ is the mean

2.96 is the mean.

Suggestions for Improvement (Highlighted part)

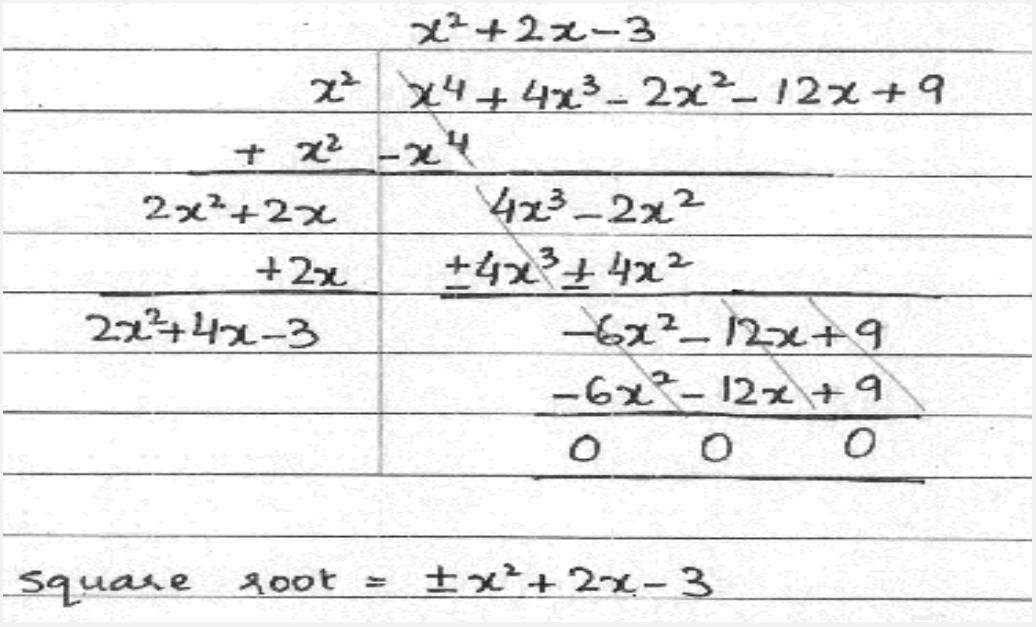
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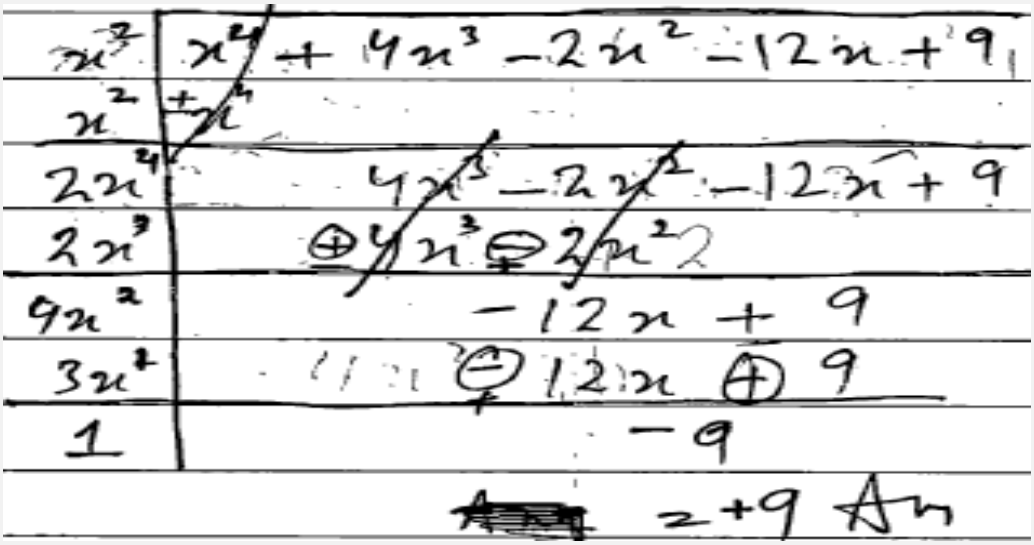
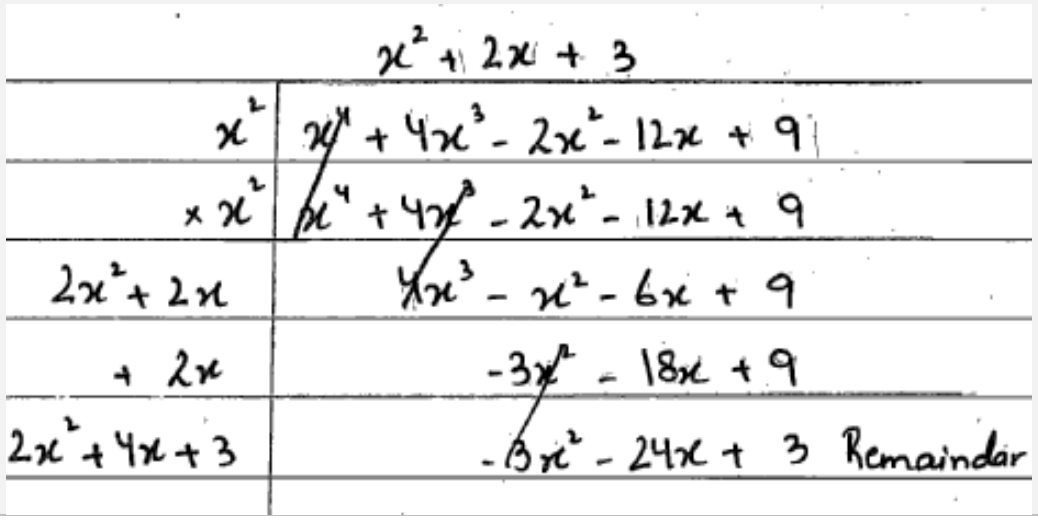
Any Additional Suggestion:

Following are teaching strategies teachers may use to address the misconceptions and errors observed in candidates' responses:

Inappropriate Completion of the Table: Emphasise the importance of accurately completing frequency distribution tables. Use examples and exercises that specifically focus on constructing these tables correctly. Provide step-by-step guidance on how to organise and calculate data within the table.

Incorrect Formula Application: Teach student to recognise the context and data type (grouped or ungrouped) in which each formula should be applied. Highlight the key differences between formulas for grouped and ungrouped data. Provide practice problems that require choosing the correct formula.
Misinterpretation of Cumulative Frequency: Teach candidates to clearly distinguish between cumulative frequency and class marks. Use visual aids and examples to illustrate the difference. Encourage students to double-check their calculations in the table before proceeding with formula application.

Question No. 3a Candidates were given the choice to attempt any ONE out of the two questions: 3a and 3b.	
Question Text	Find the square root of $x^4+4x^3-2x^2-12x+9$ by division method.
SLO No.	14.3.1
SLO Text	Calculate the square root of algebraic expression by factorisation and division.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for dividing by x^2 (1st step) 1 mark for doubling the previous divisor (2nd step) and multiplication 1 mark for the third division (3rd step) 1 mark for identifying the square root
Overall Performance	This question assessed the ability to find the square root of an algebraic expression by division method. Overall, majority of the candidates attempted this demonstrated a strong understanding of this concept and performed well.
Description of Better Responses	Better responses showed all the essential steps correctly executed. They performed the multiplications and subtractions correctly and obtained the correct result. In the end, they separately mentioned the square root of the expression.
Image of Better Response	
Description of Weaker Responses	Weaker responses exhibited candidates' challenges in forming the expressions required for multiplication, such responses made errors in simplifications. Which showed their lack of understanding of the division concept.

Images of Weaker Responses	Image (i)  Image (ii) 

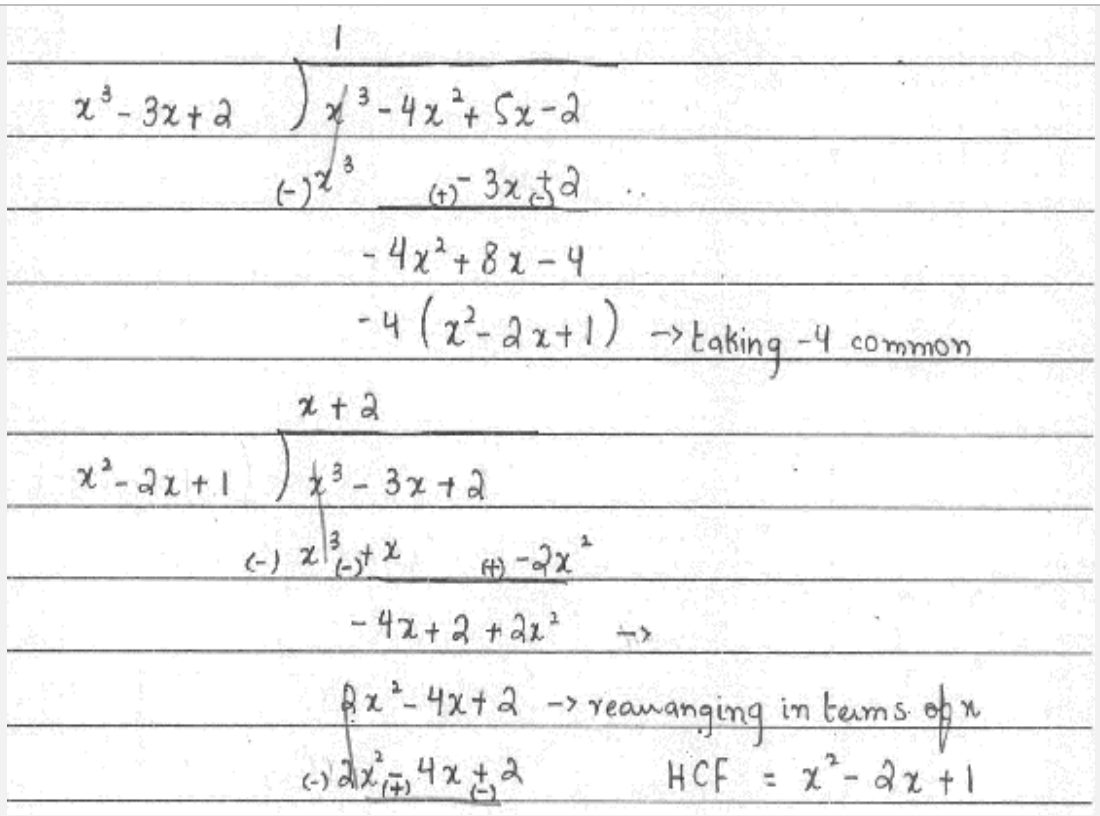
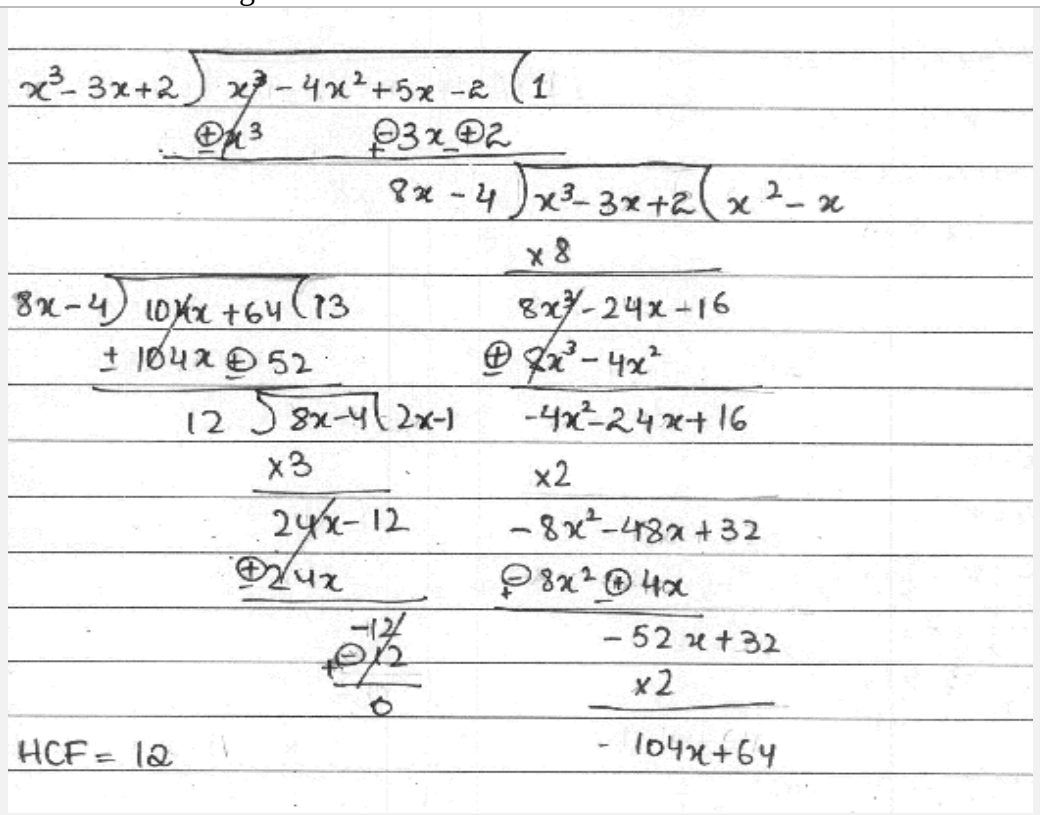
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• Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources		

Any Additional Suggestion:
Teachers may implement these teaching methodologies to develop a stronger foundation in this concept.
Step-by-Step Demonstration: Break down the process into clear, step-by-step instructions. Show students how to factor the expression and then divide it to find the square root. Use different examples to illustrate the technique.
Technology Tools: Use algebraic calculators or software like wolframalpha and mathportal.org to demonstrate how to calculate square roots of algebraic expressions. Students can then practice with similar problems.
Error Analysis: Provide examples with common mistakes and ask students to identify and correct the errors. This helps them develop problem-solving skills and error recognition.

Question No. 3b Candidates were given the choice to attempt any ONE out of the two questions: 3a and 3b.	
Question Text	Find the HCF of x^3-3x+2 and x^3-4x^2+5x-2 by division method.
SLO No.	14.1.2
SLO Text	Find the highest common factor (HCF) and the least common multiple (LCM) of algebraic expressions using division method.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for applying the division method 1 mark for first step of division correctly 1 mark for second step of division correctly after dropping off -4 1 mark for identifying the HCF from the division
Overall Performance	This question assessed the ability to find the HCF of two algebraic expressions by division method. Only a few candidates correctly attempted 3b and obtained full marks.
Description of Better Responses	Better responses to this question executed all the essential steps accurately and explicitly identified the Highest Common Factor (HCF). These candidates not only demonstrated a clear understanding of the question but also showcased a strong command of the division method.

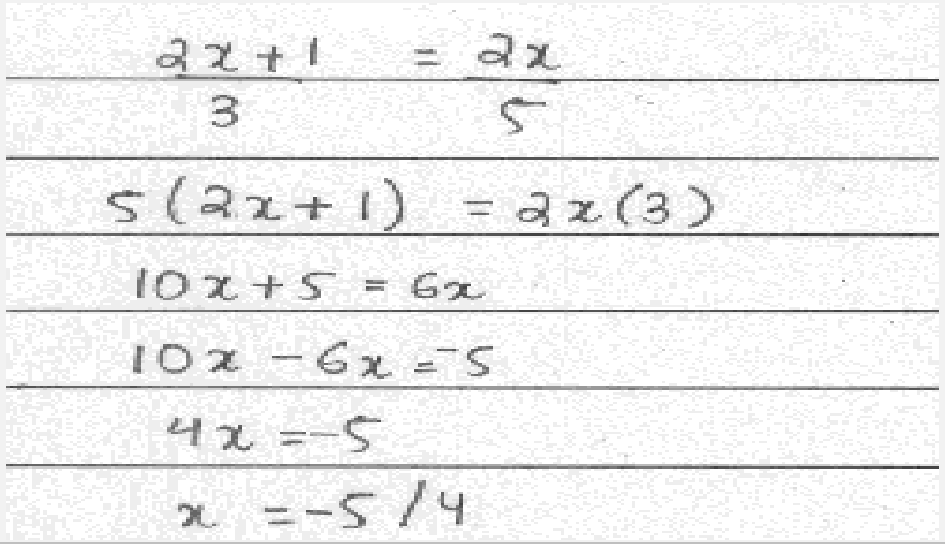
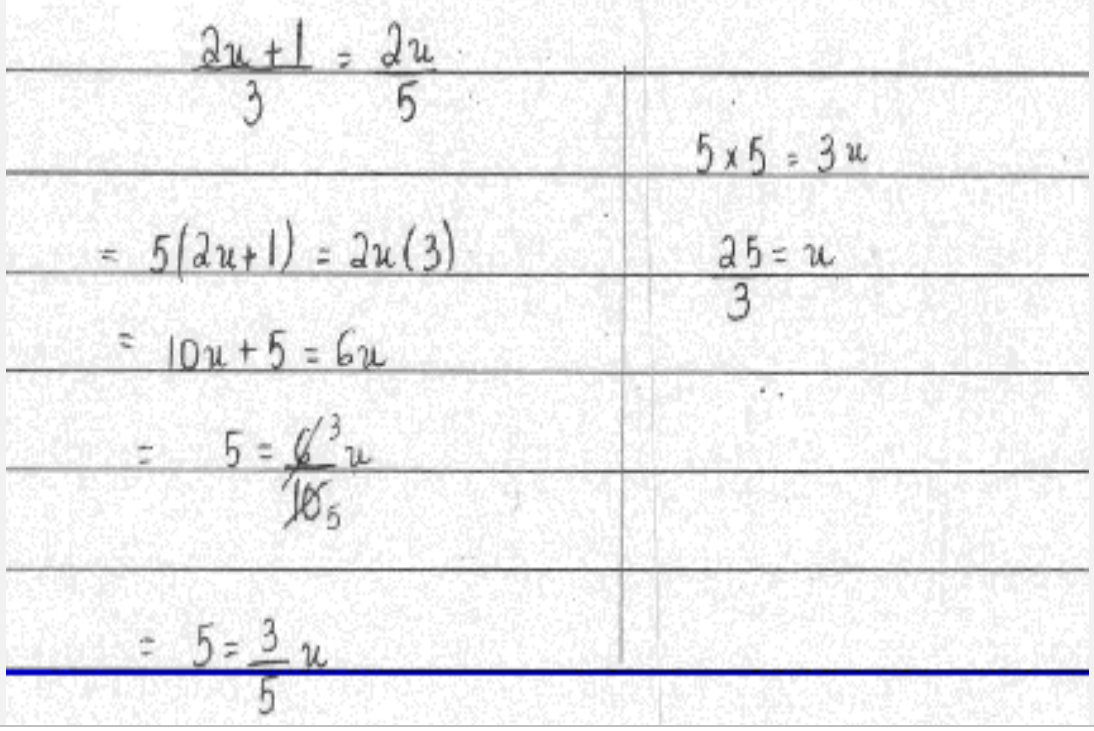
Image of Better Response	
Description of Weaker Responses	Weaker responses revealed candidates' misconceptions about the Highest Common Factor (HCF) and procedural errors. While many candidates began correctly, they subsequently made operational mistakes, particularly in subtracting or adding algebraic expressions. These candidates struggled to identify the HCF accurately, reflecting some difficulties in their understanding of the division method.
Image of Weaker Response	

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:
Teachers may implement these teaching methodologies to develop a stronger foundation in this concept.
Conceptual Clarity: Start with a strong emphasis on understanding the underlying concepts of HCF and its relationship to algebraic expressions. Ensure that students grasp the definition and significance of HCF.
Step-by-Step Approach: Teach a systematic step-by-step approach to finding the HCF. Emphasise that the division method involves dividing both expressions and finding the common factors at each step.
Error Analysis: Encourage students to analyse their mistakes. When they make errors in adding or subtracting algebraic expressions, guide them in reviewing their work to identify where they went wrong.

Question No. 4	
Question Text	Solve the equation $\frac{2x+1}{3}=\frac{2x}{5}$.
SLO No.	15.1.2
SLO Text	Solve linear equation with real coefficients.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for cross multiplication 1 mark for arranging simplification by applying addition and subtraction 1 mark for dividing by 4 on both sides to get the value of x
Overall Performance	Candidates demonstrated strong performance in this question, which cantered on solving a linear equation in one variable. The majority of candidates achieved full marks, highlighting their clear understanding of linear equations.
Description of Better Responses	Better responses demonstrated that candidates had no difficulty clearing the given equation. Such responses correctly followed each step of the method. Initially, they rearranged the equation in a proper way, and then they used cross multiplication to

	simplify it. This careful and accurate approach ultimately led them to the correct value of *x*.
Image of Better Response	
Description of Weaker Responses	In weaker responses, candidates encountered challenges when attempting to simplify the linear equation. While they correctly initiated the process with cross multiplication and expanded it accurately, they encountered difficulty in isolating the value of *x*.
Image of Weaker Response	

Suggestions for Improvement (Highlighted part)

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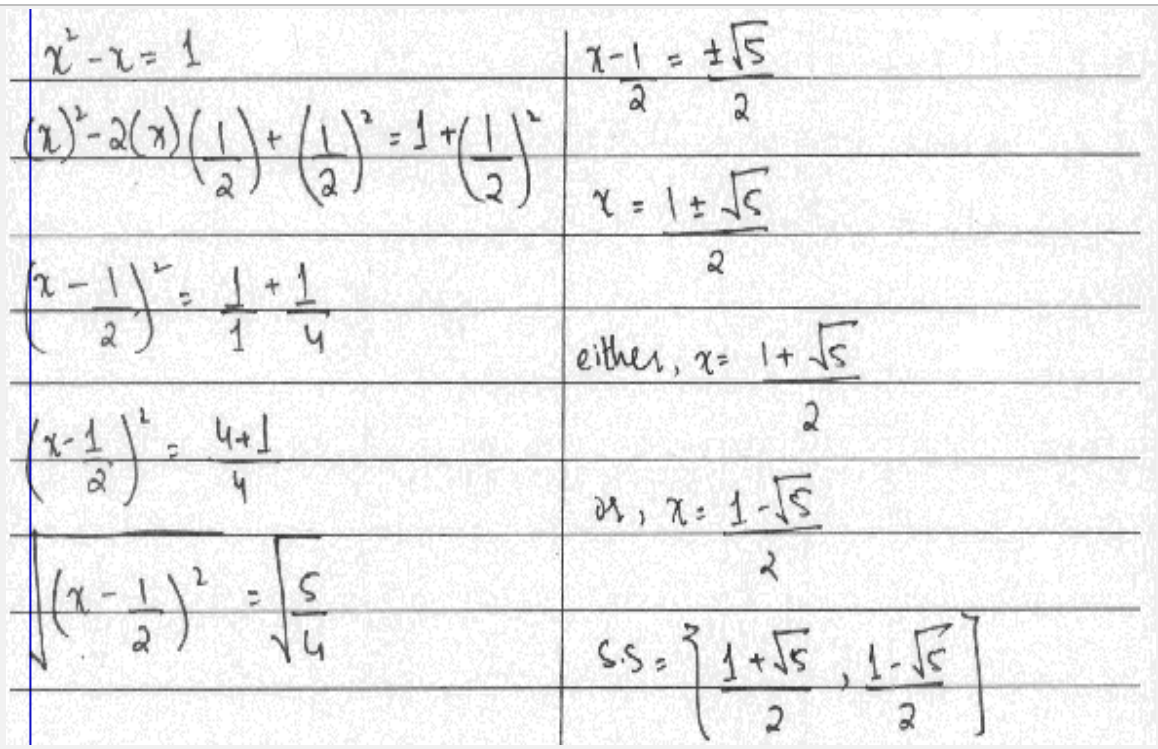
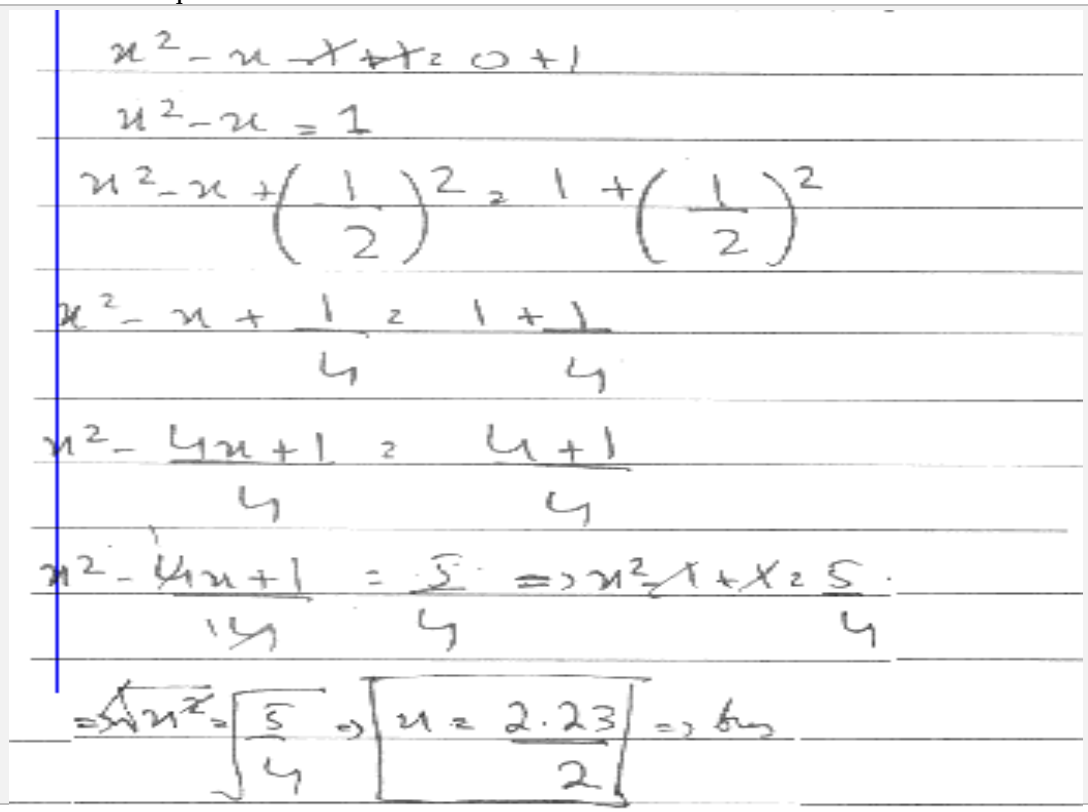
Any Additional Suggestion:

Step-by-Step Demonstrations: Break down the solving process into clear and simple steps. Use color coding or highlighters to emphasise key components, such as variables, constants, and operations.

Interactive Online Tools: Explore online equation-solving tools or apps that allow students to input equations and see step-by-step solutions. Websites like Desmos or Wolfram Alpha can be helpful. Utilise digital tools like graphic calculators, interactive simulations, or educational apps to demonstrate and solve linear equations with visuals and animations.

Provide video tutorials or online resources for students to review before class, allowing class time to be dedicated to solving equations together and addressing questions.

Question No. 5	
Question Text	Solve the equation $x^2 - x - 1 = 0$ by completing square method.
SLO No.	16.2.1b
SLO Text	Solve a quadratic equation in one variable by b) completing square method.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for rearranging and taking 1 on the RHS 1 mark for identifying and adding the square of ½ on both sides 1 mark for making complete square 1 mark for taking square root
Overall Performance	This question evaluated candidates' ability to solve a quadratic equation using the completing the square method. However, candidates faced significant challenges with this question, fewer of the candidates were able to achieve full marks.
Description of Better Responses	Better responses showed the correct working required for the solution. Candidates showing such responses identified the term to be added on the left hand side of the equation to make the given expression a complete square. They took the square root on both sides and obtained the correct values of *x*. Hence, such responses scored full marks.

Image of Better Response	
Description of Weaker Responses	Weaker responses, highlighted significant challenges. Some candidates either could not identify the appropriate method to solve the question or made procedural errors when applying the correct method. In particular, those who applied the correct method, had difficulty in forming the necessary expression to complete the square, and they also struggled with the decision of whether to add or subtract this expression on the left or right side of the equation.
Image of Weaker Response	

Suggestions for Improvement (Highlighted part)

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Any Additional Suggestion:
Teachers may implement these teaching methodologies to develop a stronger foundation in this concept.
Concrete Examples and Visual Aids: Start with concrete examples where students can physically manipulate objects or tiles to form a square. Use visual aids like square tiles or diagrams to represent the process of completing the square.
Geometric Interpretation: Emphasise the geometric interpretation of completing the square. Show how the process relates to creating a perfect square trinomial, and how it transforms a quadratic equation into a square form.
Error Analysis: Present common mistakes that students make, such as adding or subtracting the wrong value when completing the square. Encourage students to analyse these errors and explain why they are incorrect.

Question No. 6	
Question Text	The 5th term and 6th term of an arithmetic sequence is 30 and 35 respectively. Calculate the i. common difference. ii. 1st term of the sequence. iii. value of n, if the nth term is 55.
SLO No.	17.2.2
SLO Text	Solve problems involving arithmetic sequence.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for calculating d ii. 1 mark for calculating the 1st term iii. 1 mark for writing $55 = 10 + (n-1)5$ 1 mark for making calculating n

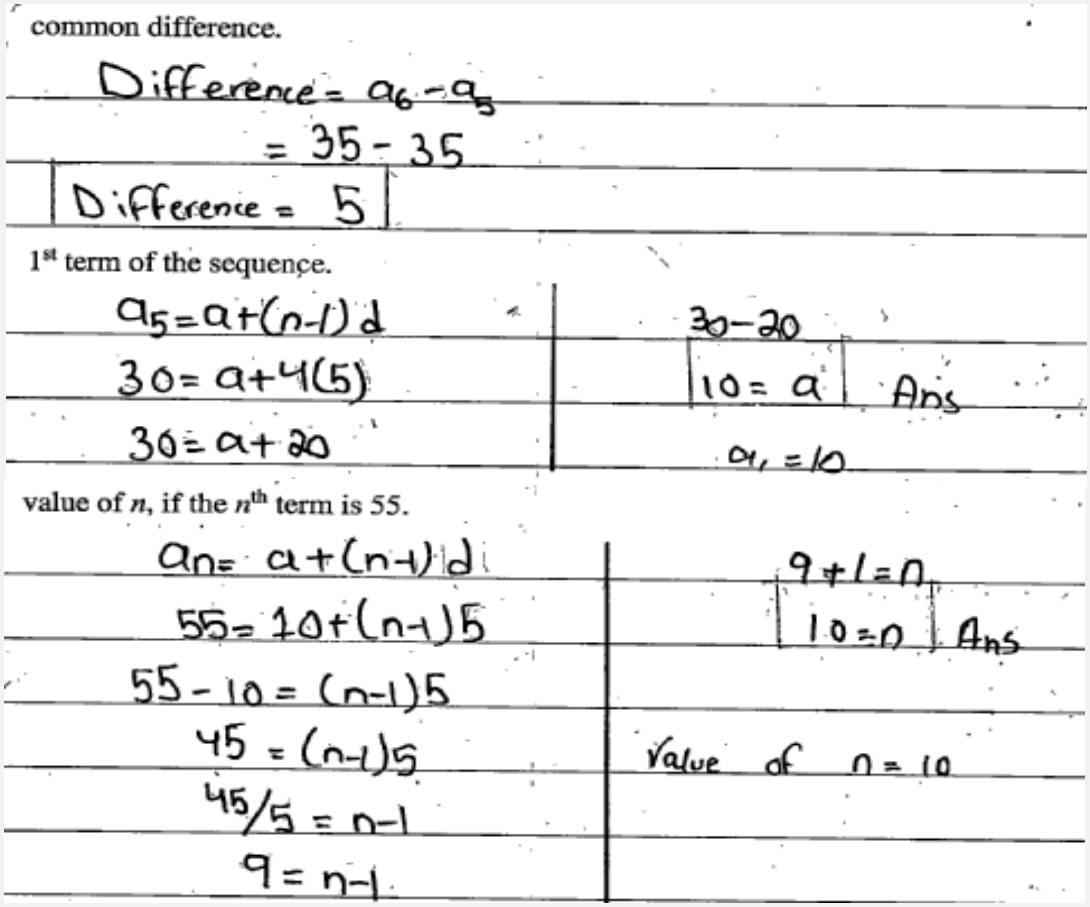
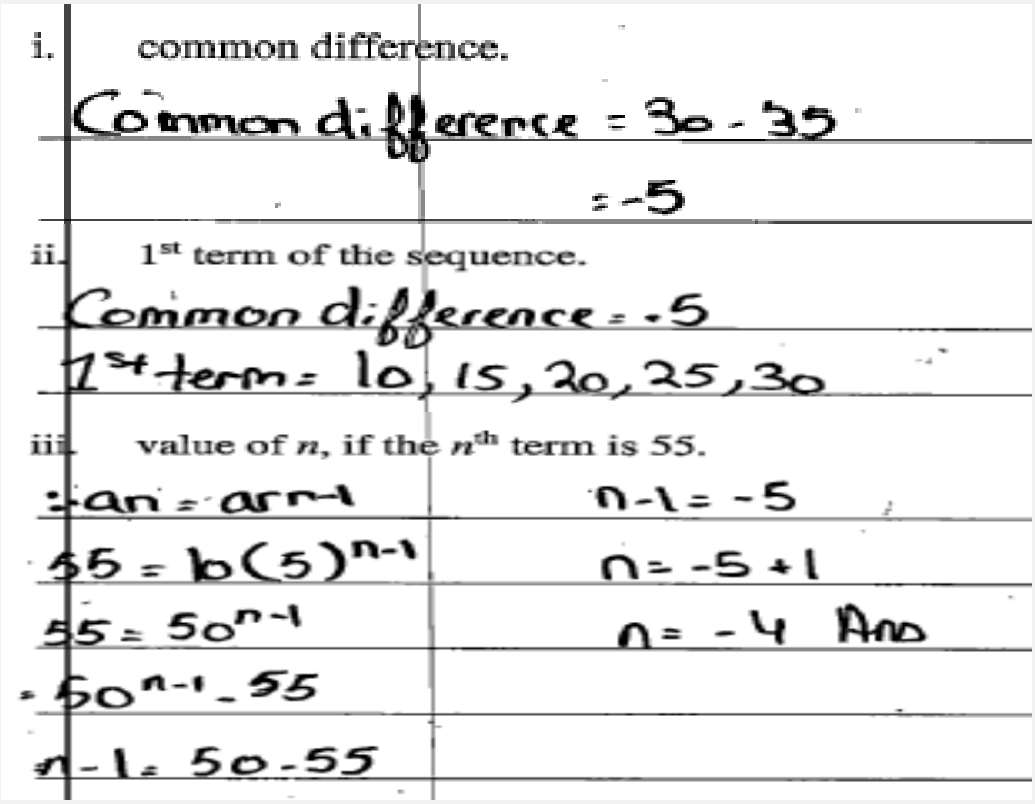
Overall Performance	This question consisted of three parts, assessing candidates' proficiency in solving problems related to arithmetic progression. Overall, a substantial number of candidates secured full marks. Most candidates successfully completed part (i) of the question, while some did not attempt part (ii). In part (iii), only a minority of candidates were able to earn marks.
Description of Better Responses	The stronger responses in this question showcased a clear grasp of arithmetic progression and its fundamental concepts, such as the common difference, the first term, and the nth term. Such candidates accurately determined these components through precise calculations. Some candidates achieved full scores by effectively utilising the common difference and constructing the series without relying on specific formulae.
Image of Better Response	
Description of Weaker Responses	In weaker responses, though candidates were able to find the common difference correctly, however, they were unable to apply the nth term formula appropriately. Specifically, candidates could not identify the values and substitute them in corresponding expressions. i.e., candidates substituted 55 in *n* instead of a_n and evaluated the incorrect value.

Image of Weaker Response	

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

Following are the teaching strategies that teachers may use to teach arithmetic sequence:

Number Line: Draw a number line on the board and start with an initial value. Show students how to generate an arithmetic sequence by adding the common difference at regular intervals. This visual representation can be powerful. Show students how to graph arithmetic sequences on a coordinate plane. This can help them visualise the linear nature of these sequences.

Comparing Sequences: Present multiple arithmetic sequences with different starting values and common differences. Have students compare and contrast these sequences to understand how they differ.

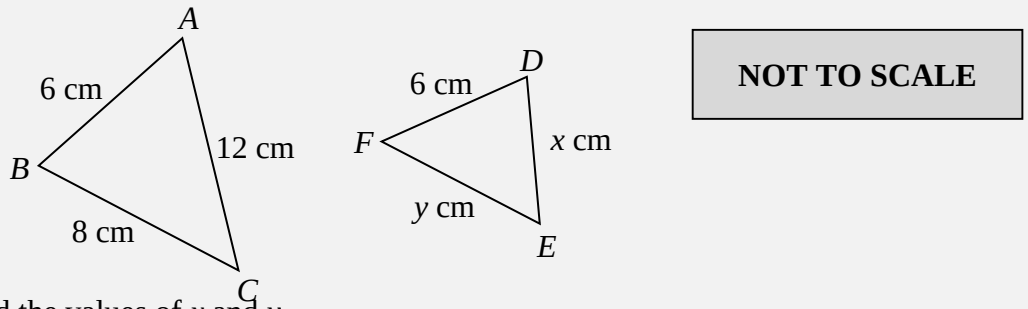
Question No. 7a Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b.	
Question Text	In the correspondence $\Delta ABC \leftrightarrow \Delta DEF$, ABC and DEF are similar.  Find the values of x and y.
SLO No.	18.3.2f
SLO Text	Apply the following properties of congruency and similarity of triangles to solve related problems: if two triangles are similar, the measures of their corresponding sides are proportional.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for identifying the corresponding sides. 1 mark for taking ratio of corresponding sides as proportional i.e. $\frac{AB}{DE}=\frac{AC}{DF}$ and $\frac{AC}{DF}=\frac{BC}{EF}$. 1 mark for calculating x. 1 mark for calculating y.
Overall Performance	Majority of the candidates attempted this question, but only some of the candidates were able to obtain full marks. Candidates exhibited confusion in differentiating congruence and similarity, taking ratios of corresponding sides and solving ratios.
Description of Better Responses	Better response exhibited sufficient understanding of correspondence given as $\triangle ABC \leftrightarrow \triangle DEF$ and similar triangles. Moreover, they applied the concept of taking ratio of corresponding sides and equated to solve for the unknown such as $\frac{AB}{DE}=\frac{BC}{EF}=\frac{AC}{DF}$

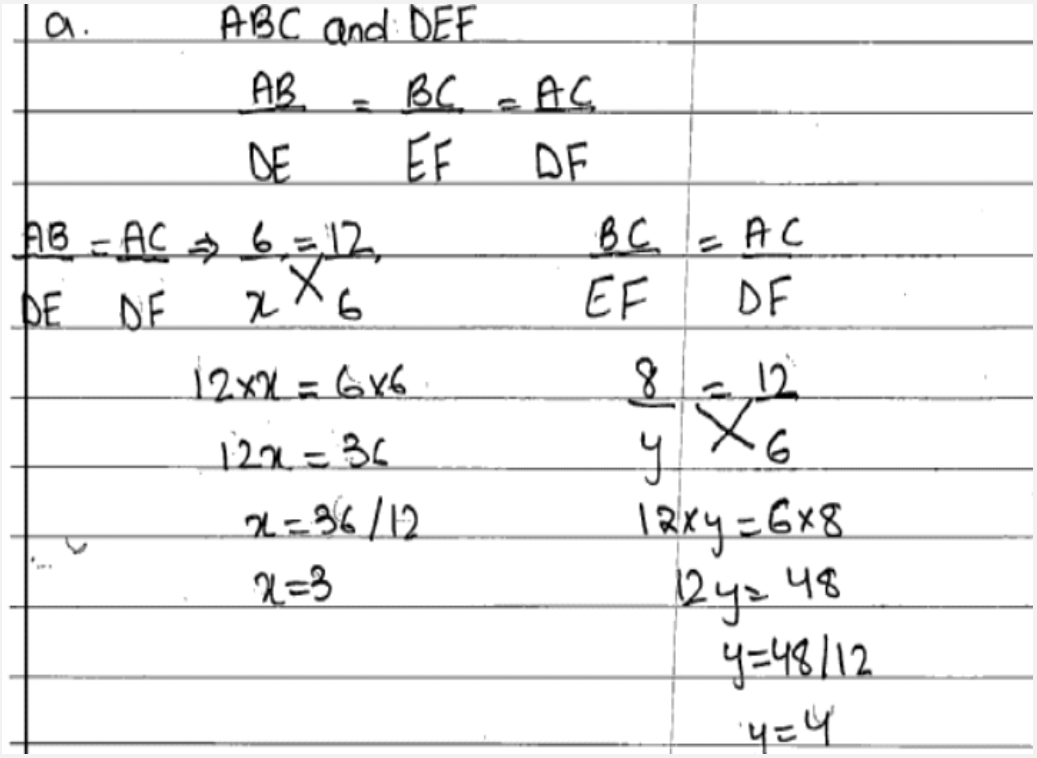
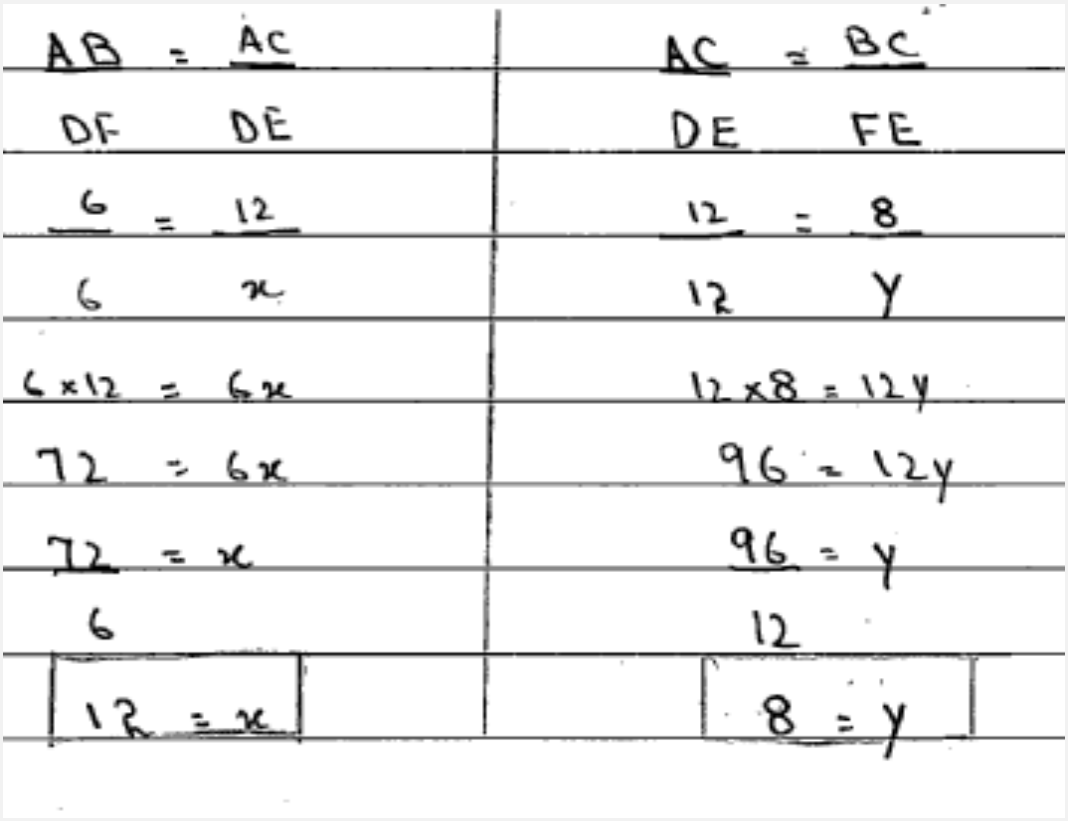
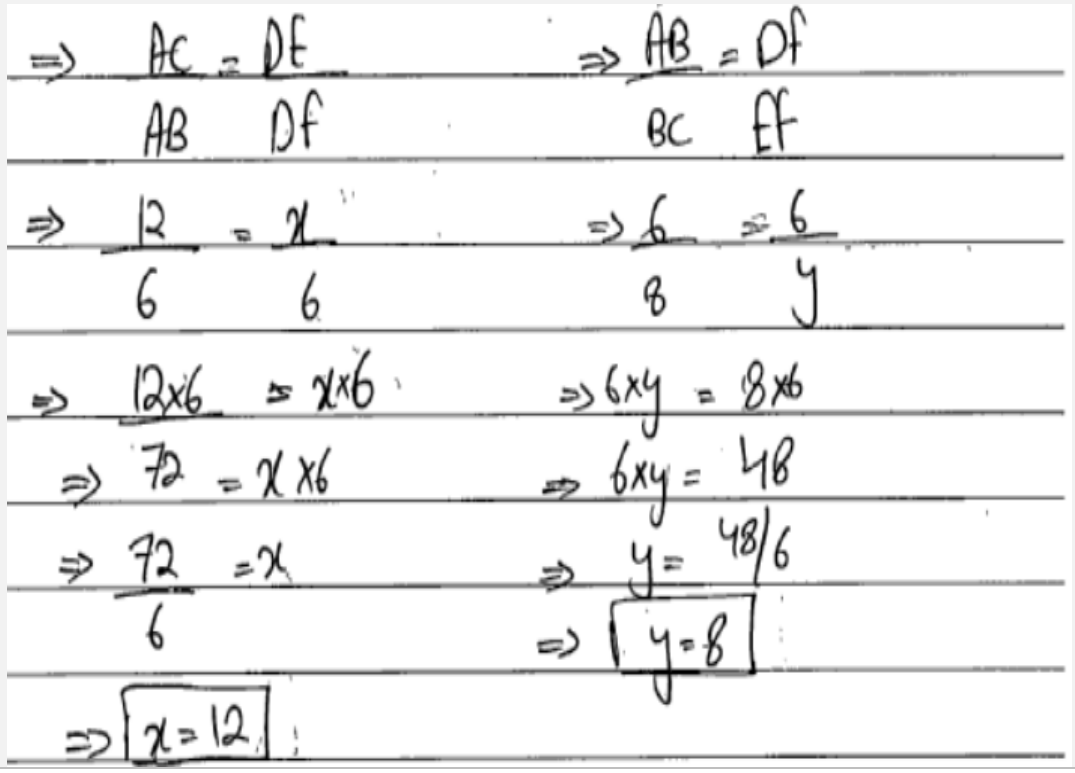
Image of Better Response	
Description of Weaker Responses	Weaker response exhibited different conceptual gaps. Firstly, some candidates were unable to identify the corresponding sides, hence could not take the correct ratios. Secondly, candidates equated incorrect ratios. Thirdly, candidates also demonstrated calculation errors in re-arranging and solving for the unknowns.
Images of Weaker Responses	**Image (i)** 

Image (ii)



Suggestions for Improvement (Highlighted part)

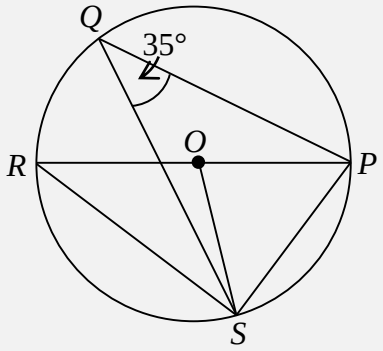
How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:
Following are the teaching strategies to address the misconceptions and errors observed in responses.
Identifying Corresponding Sides:
Visual Aids: Use visual aids like diagrams and geometric figures to clearly illustrate the corresponding sides in similar triangles. Emphasise the importance of visual identification before applying ratios.

Practice Matching Sides: Provide exercises where students must match the corresponding sides of similar triangles before proceeding to calculate ratios. This helps reinforce the concept.
Real-Life Analogies: Draw analogies to real-life scenarios where recognising corresponding elements is crucial. For example, comparing similar triangles to maps or architectural blueprints where proportions must be accurately preserved.
Equating Ratios Correctly: Common Mistakes Highlight: Share common mistakes students make when equating ratios, and provide examples where ratios are equated incorrectly to illustrate the consequences.
Step-By-Step Approach: Teach a step-by-step method for setting up and solving proportions. Emphasise that corresponding sides must have the same ratio.
Calculator Skills: Spend time teaching students how to use calculators effectively to perform arithmetic calculations. Emphasise the importance of double-checking entries.
Check Answers: Encourage students to verify their solutions by plugging their results back into the original problem to see if they hold true.
Error Analysis: Present worked examples with intentional calculation errors and ask students to identify and correct these errors. This helps them become more vigilant in their calculations.
Estimation: Teach estimation techniques to help students identify whether their answers are reasonable.

Question No. 7b

Candidates were given the choice to attempt any ONE out of the two questions: 7a and 7b.

Question Text	In the given diagram, a circle with centre O is given.  **NOT TO SCALE** If the measure of the angle SQP is 35°, find the angle i. SOP. ii. SRP. iii. SOR. iv. RPS.
SLO No.	18.5.2a, 18.5.2b and 18.5.2c
SLO Text	Apply the following properties of a circle to find the unknown measurements a) the angle in a semicircle is a right angle b) the angles in the same segment of a circle are equal c) the central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.
Max Marks	4
Cognitive Level	A

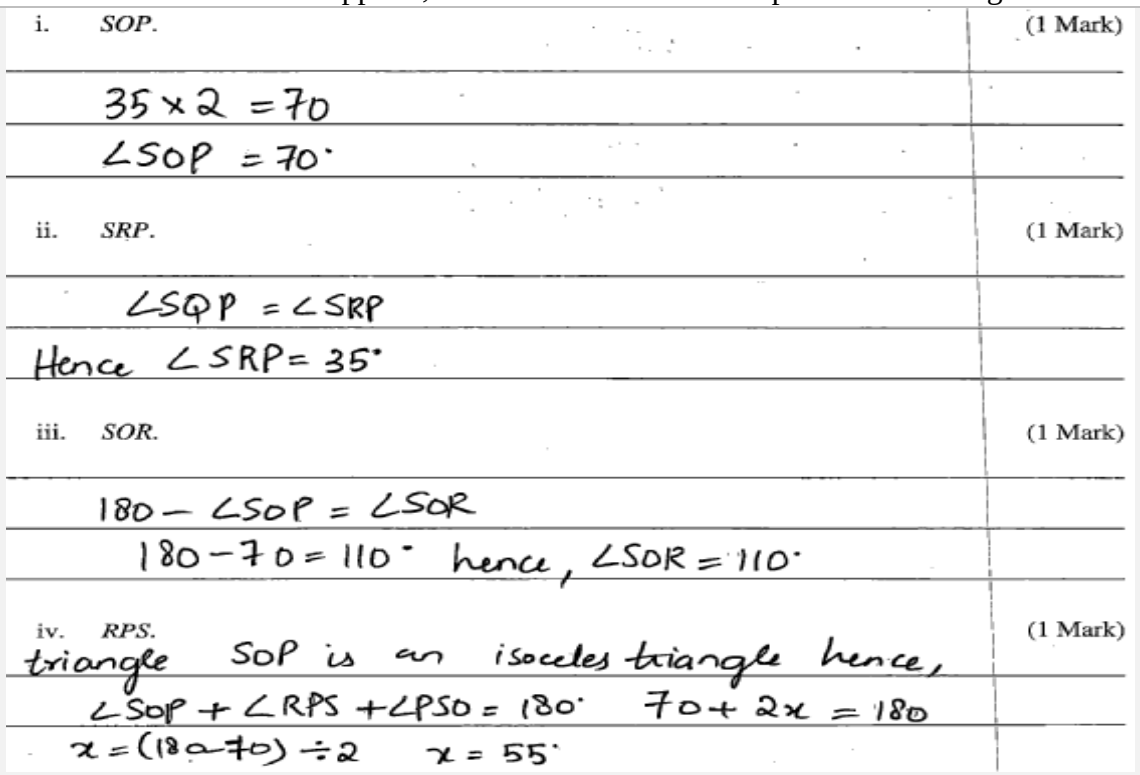
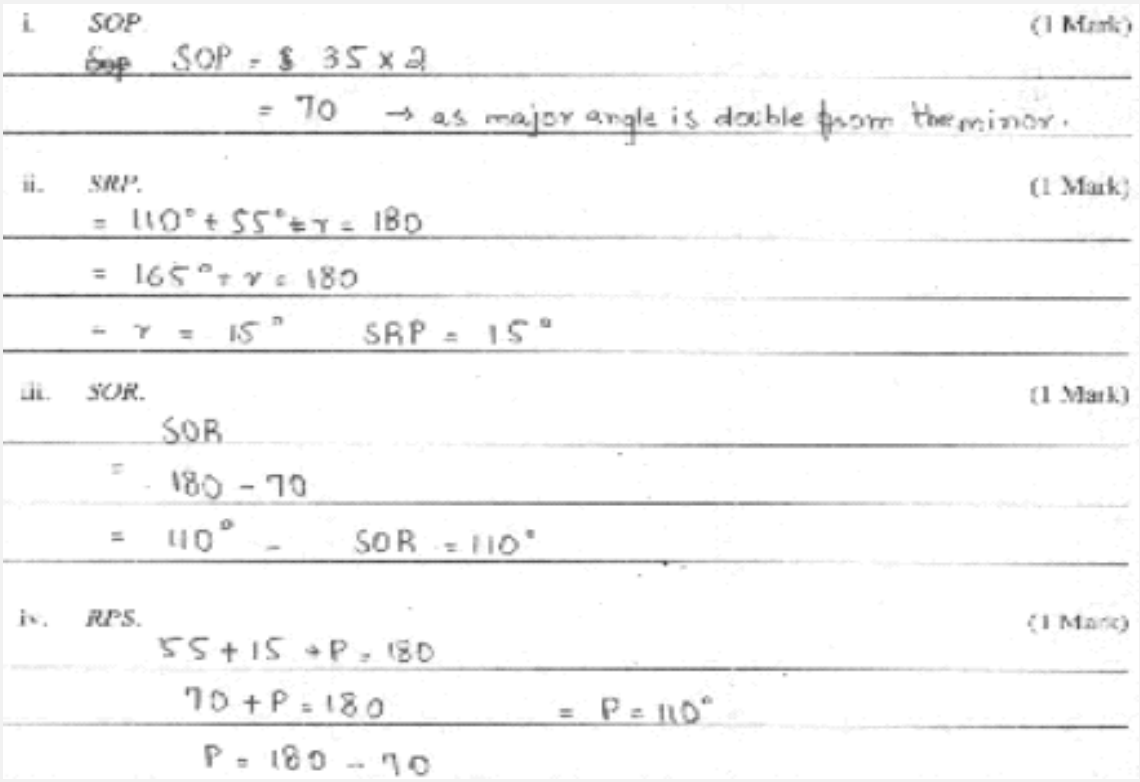
Checking Hints	1 mark for calculating $\angle SOP$ 1 mark for calculating $\angle SQP$ 1 mark for calculating $\angle SOR$ 1 mark for calculating $\angle RPS$
Overall Performance	This question assessed candidates' understanding of circle properties. It was observed that a relatively small number of candidates attempted this question, and among those who did, only a few candidates were able to secure full marks.
Description of Better Responses	Better responses reflected sufficient understanding of the concepts related to circle properties and theorems. Candidates demonstrating better responses analysed the given diagram in view of the related theorems and executed correct calculations to evaluate the unknown angles namely *SRP*, *SOR* and *RPS*. Although the question did not require reason or references of the related theorems, however, some candidates gave reasons and reference of the theorem applied; which hence reflected deeper understanding.
Image of Better Response	
Description of Weaker Responses	Weaker responses reflected insufficient understanding and application of the circle properties. In attempting this question, the candidates could not analyse the given diagram in view of the related theorem, they randomly took half or double of the angles shown in the diagram. In addition, some candidates made suppositions instead of using the given information in the question. i.e., some candidates assumed triangle *OSP* as right angled triangle. In addition, some took *OST* as right angle despite of the fact that the figure was not to scale.

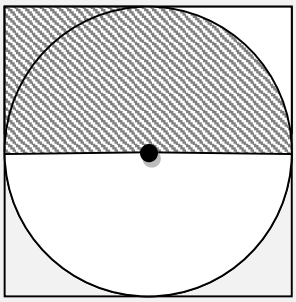
Image of Weaker Response	

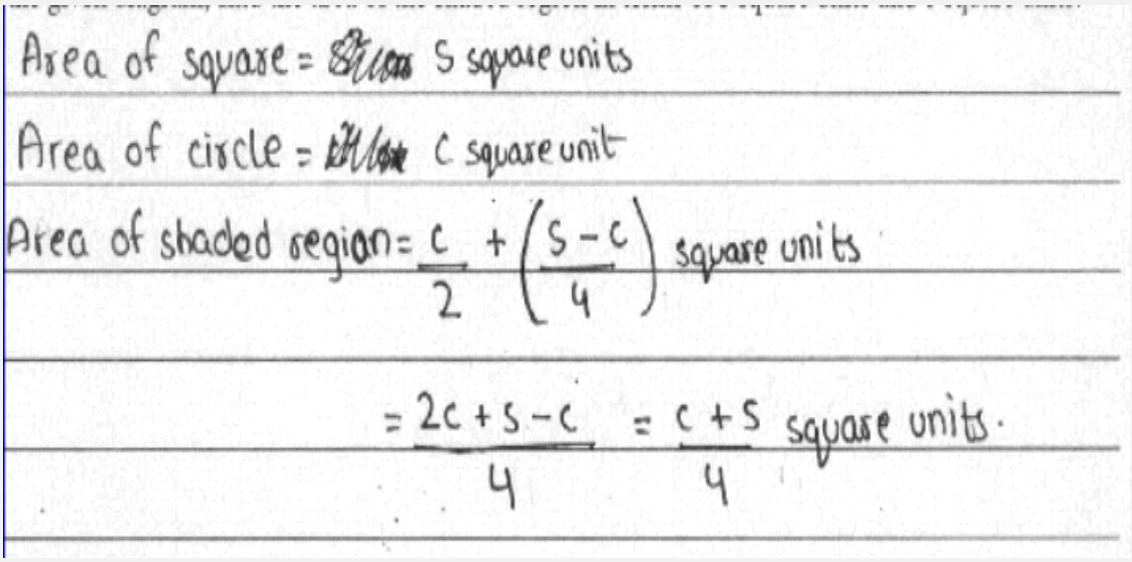
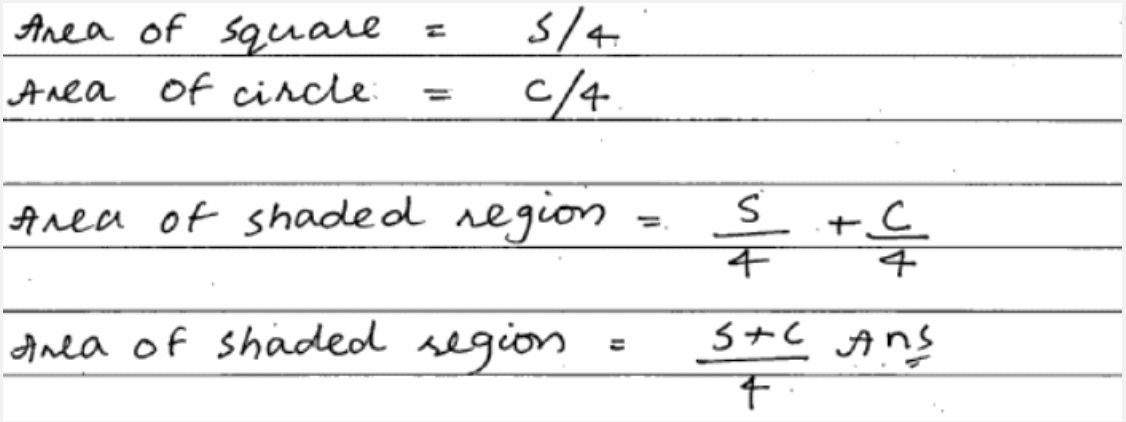
Suggestions for Improvement (Highlighted part)

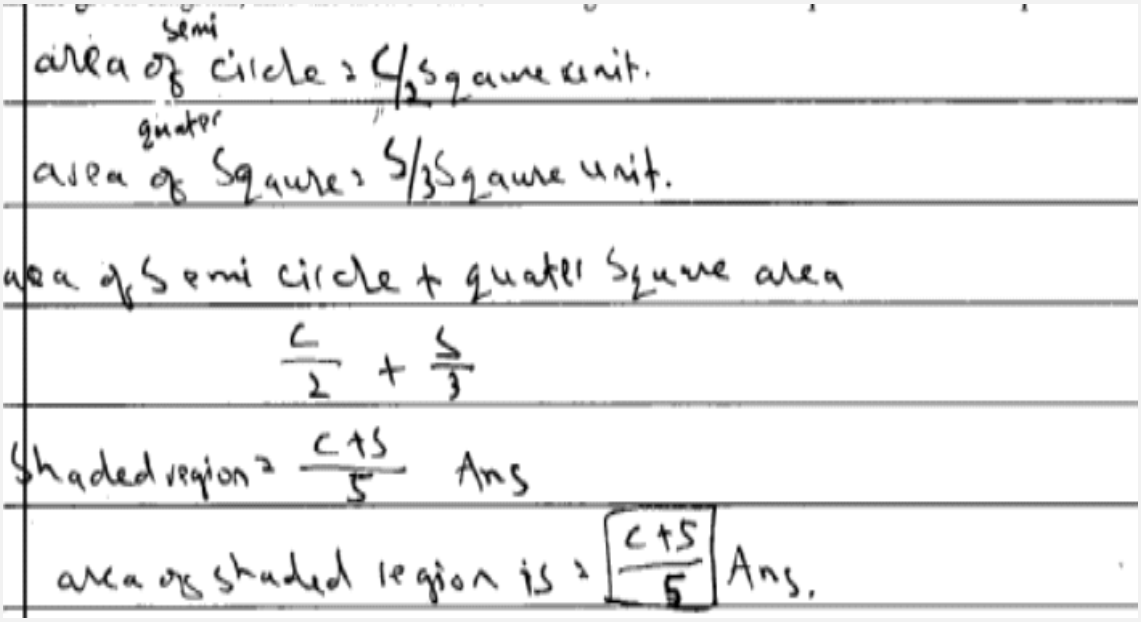
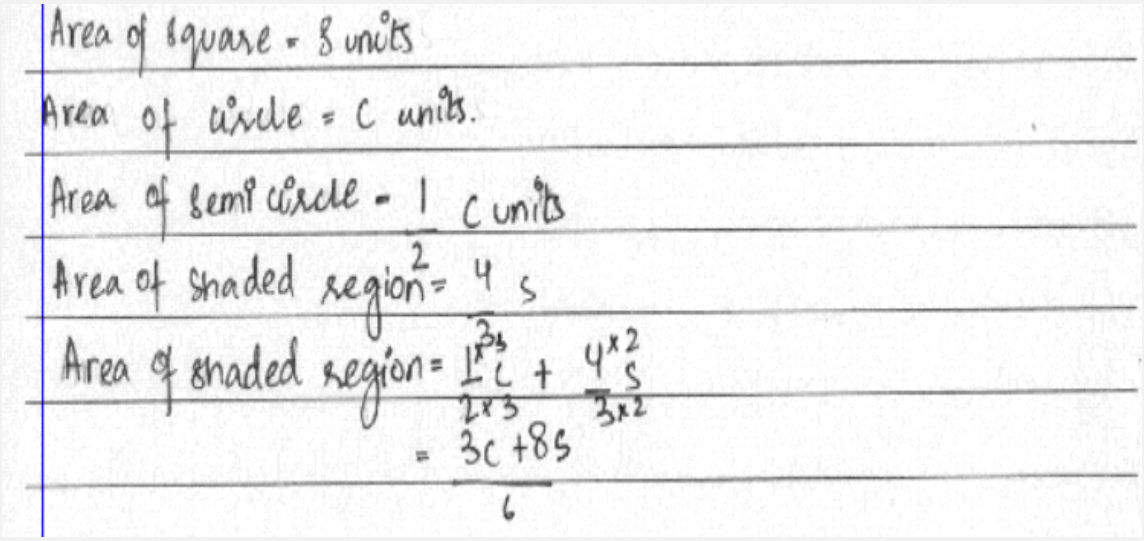
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Any Additional Suggestion:
Teachers may implement these strategies to improve students' performance on this concept.
Clear Conceptual Foundation: Start by building a strong foundation in the basic properties of circles, such as radius, diameter, circumference, and area. Ensure that students have a clear understanding of these fundamental concepts.

Geometric Visualisation: Use visual aids like drawings, diagrams, and interactive software to help students visualize circle properties. Show how these properties relate to the shape of a circle.
Practical Applications: Discuss real-life applications of circle properties, such as calculating the area of circular objects, designing wheels, or understanding orbits in astronomy. This can make the topic more relatable and engaging.
Visual Proofs: Explore visual proofs of circle properties. For instance, use a circle's diameter to show that it divides the circle into two congruent semicircles.

Question No. 8	
Question Text	In the given diagram, the area of the square is *s* square units and the area of the circle is *c* square units.  **NOT TO SCALE** In the given diagram, find the area of the shaded region in terms of *s* square units and *c* square units.
SLO No.	19.2.4
SLO Text	Solve problems related with areas of triangle, rectangle, square, parallelogram and circle.
Max Marks	4
Cognitive Level	A
Checking Hints	**Method I:** 1 mark for finding the area of the semi-circle 1 mark for finding the area of the square which is not covered by the given circle 1 mark for finding the area of shaded corner 1 mark for finding the required shaded area **Method II:** 1 mark for splitting the shaded region into quarter of the circle and quarter square (**Note**: mark would be granted even if a candidate does not show on the diagram.) 1 mark for finding the area of the quarter square as $\frac{1}{4}s$ square units 1 mark for finding the area of the quarter circle as $\frac{1}{4}c$ square units 1 mark for adding the two shaded parts together as $\frac{1}{4}s+\frac{1}{4}c$ *OR* $\frac{s+c}{4}$ square units
Overall Performance	Overall, this question proved to be challenging Majority of the students opted to solve the question using the formulae for finding the area of the square and the circle. However, the question demanded to calculate a selected area based on given areas of the square and the circle.

Description of Better Responses	Better responses reflected strong logical reasoning to solve this question. Candidates used different methods to solve this question. Firstly, some of the candidates solved this question by splitting the figure into a semicircle and square; then added their area as $\frac{s}{4}+\frac{c}{4}$. Secondly, candidates subtracted $\frac{s-c}{4}$ from the area of the semi-circle $\frac{c}{2}$.
Images of Better Responses	**Image (i)**  Area of square = S square units Area of circle = C square unit Area of shaded region = $\frac{C}{2}+\left(\frac{S-C}{4}\right)$ square units $=\frac{2C+S-C}{4}=\frac{C+S}{4}$ square units. **Image (ii)**  Area of square = S/4 Area of circle = C/4 Area of shaded region = $\frac{S}{4}+\frac{C}{4}$ Area of shaded region = $\frac{S+C}{4}$ Ans
Description of Weaker Responses	Weaker responses reflect candidates' difficulty in identifying the shaded portion as a combination of area of the square and circle. Some candidates connected the formula for area of a circle and square and tried to come up with a new formula. Hence, they could not reach the required result.

Images of Weaker Responses	**Image (i)**  area of semi circle = C/2 square unit. area of quater Square = S/3 Square unit. area of Semi circle + quater Square area C/2 + S/3 Shaded region = (C+S)/5 Ans area of shaded region is = (C+S)/5 Ans. **Image (ii)**  Area of Square = 8 units Area of circle = C units. Area of Semi circle = 1/2 C units Area of shaded region = 4/2 S Area of shaded region = 1×3C/2×3 + 4×2S/3×2 = 3C+8S/6

Suggestions for Improvement (Highlighted part)

How to Approach SLO	Pedagogy Used for that SLO	Assessment Strategies
• Understand the expectations of the command words • Look at the cognitive level • Identify the content that is required to answer that question (both in terms of understanding of concepts and any skills	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual resources • Think, Pair and Share • Questioning Technique (Socratic approach)	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

that may be required like analysing or evaluating) • Go through the past paper questions on that particular concept • Refer to the resource guide for extra resources	• Practical Demonstration	

Additional Suggestions:

Teachers may implement these strategies to improve students' performance on this concept.

Clarify the Question: Train students to read and understand the question carefully. Ensure they grasp the specific requirement of the problem, such as calculating the shaded area.

Error Analysis: Share examples of common mistakes, such as using incorrect formulas or adding areas that should be subtracted. Ask students to identify and correct these errors.

Encourage Critical Thinking: Pose questions that require critical thinking, such as asking students to explain why they are adding or subtracting specific areas. Encourage them to justify their steps.

Visualise the Problem: Encourage students to visualise the problem by drawing diagrams. This can help them see the relationship between the given areas and the shaded portion.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange

their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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